SOLUTION

SECTION–I

1. Ans. (A)

$\log(\sin^2 x.\cos x) = t$

$\Rightarrow (2\sin x.\cos^2 x - \sin^3 x)dx = dt.\sin^2 x.\cos x$

$\Rightarrow (2\sin x(1-\sin^2 x) - \sin^3 x)dx = dt.\sin^2 x.\cos x$

$\Rightarrow (2\sin x - 3\sin^3 x)dx = dt\dfrac{\sin x.\sin 2x}{2}$

$\Rightarrow \dfrac{(2-3\sin^2 x)}{\sin 2x}dx = \dfrac{dt}{2}$

$I = \dfrac{1}{2}\int t dt = \dfrac{t^2}{4} + C$

$= \dfrac{\log^2(\sin^2 x.\cos x)}{4} + C$

$= \log^2\left(\dfrac{1}{2}\sin 2x.\sin x\right) + C$

2. Ans. (1)

$\int \ell n\left(\sqrt{2+x} + \sqrt{2-x}\right)dx$ (By parts)

$= x\ell n\left(\sqrt{2+x} + \sqrt{2-x}\right),$

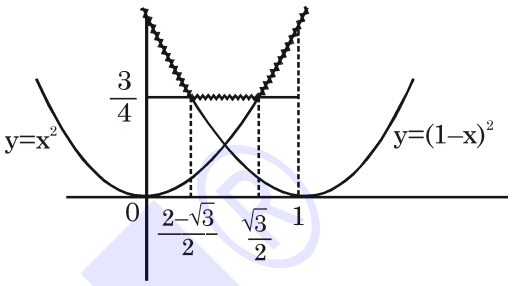
$- \int x.\dfrac{\left(\dfrac{1}{2\sqrt{2+x}} - \dfrac{1}{2\sqrt{2-x}}\right)}{\sqrt{2+x} + \sqrt{2-x}}dx$

$= x\log\left(\sqrt{2+x} + \sqrt{2-x}\right) - \dfrac{1}{4}\int \dfrac{4 - 2\sqrt{4-x^2}}{\sqrt{4-x^2}}dx$

$= xf(x) + \sin^{-1}\dfrac{x}{2} - \dfrac{x}{2} + C$

3. Ans. (A)

Sol.



$\int_0^1 f(x) = \int_0^{\frac{2-\sqrt{3}}{2}} (1-x)^2 dx + \int_{\frac{2-\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} \frac{3}{4}dx + \int_{\frac{\sqrt{3}}{2}}^{1} x^2 dx$

$= \dfrac{6\sqrt{3}-1}{12}$

4. Ans. (A)

Sol. $I = \int_{\pi/3}^{\pi/6} \ell n \sin x dx + \int_{\frac{1}{2}\ell n\frac{3}{4}}^{-\ell n 2} \sin^{-1} e^x dx$

$= -\dfrac{\pi}{6}\ell n 2 - \dfrac{\pi}{6}\ell n\dfrac{3}{4} = \dfrac{\pi}{6}\ell n\dfrac{2}{3}$

5. Ans. (A)

Sol. $\int_0^y e^t dt + \int_0^x f(t)dt = g(x)$ .....(1)

diff. w.r.t. x

$e^y\dfrac{dy}{dx} + f(x) = g'(x)$

$\Rightarrow \dfrac{dy}{dx} = \dfrac{g'(x) - f(x)}{e^y}$ .....(2)

Again (1)

$\Rightarrow e^y - 1 = g(x) - \int_0^x f(t)dt$

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$$\Rightarrow \quad e^y = 1 + g(x) - \int_0^x f(t)dt$$

$$\therefore \quad (2) \quad \Rightarrow \quad \frac{dy}{dx} = \frac{g'(x) - f(x)}{1 + g(x) - \int_0^x f(t)dt}$$

6. Ans. (A)

$$\int e^x \left(\frac{1}{\ell nx} - \frac{1}{(x)(\ell nx)^2} \right) dx$$

$$\int \frac{e^x}{\ell nx} dx - \int \frac{e^x}{x(\ell nx)^2} dx$$

$$\left[\frac{1}{\ell nx} e^x - \int \frac{-1}{(\ell nx)^2 x} e^x dx \right] - \int \frac{e^x}{x(\ell nx)^2} dx + C$$

(Applying integration by parts)

$$= \frac{e^x}{\ell nx} + C$$

7. Ans. (A)

$$\lim_{x \to 0} \frac{\ell n(1+x) + \frac{x}{1+x}}{f(x).\sin x} = 5$$

$$\lim_{x \to 0} \frac{\frac{\ell n(1+x)}{x} + \frac{1}{1+x}}{f(x).\frac{\sin x}{x}} = 5$$

$$\frac{2}{5} = f(0)$$

8. Ans. (A)

$$I_2 = \int_0^1 \frac{x^2}{e^{x^3}(2 - x^3)} dx = \frac{1}{3} \int_0^1 \frac{dz}{e^{1-z}(z+1)}, (z = 1 - x^3)$$

$$= \frac{1}{3} \int_0^1 \frac{e^z}{e(z+1)} dz = \frac{1}{3e} I_1$$

$$\frac{I_1}{I_2} = 3e$$

9. Ans. (A,C,D)

$$\int \frac{\sec^2 x + 3}{\sqrt{\sec^2 x + 3}} dx = \int \frac{\sec^2 x dx}{\sqrt{4 + \tan^2 x}} + \int \frac{3 \cos x}{\sqrt{4 - 3\sin^2 x}} dx$$

$$= \int \frac{dt}{\sqrt{4 + t^2}} + \int \frac{\sqrt{3}\, d\theta}{\sqrt{4 - \theta^2}}$$

$$= \ell n \left| t + \sqrt{4 + t^2} \right| + \sqrt{3} \sin^{-1} \left(\frac{\theta}{2} \right) + C$$

$$= \ell n \left| \tan x + \sqrt{4 + \tan^2 x} \right| + \sqrt{3} \sin^{-1} \left(\frac{\sqrt{3} \sin x}{2} \right) + C$$

$f(x) = \tan x$, $g(x) = \sin x$,

$k = \sqrt{3}$

10. Ans. (A,B,C,D)

Let $I = \int (2x^6 + 15x^4 + 2x^2 + 3) \cos 2x \, dx$

$= \int (2x^6 + 2x^2).\cos 2x + \int (15x^4 + 3).\cos 2x dx$

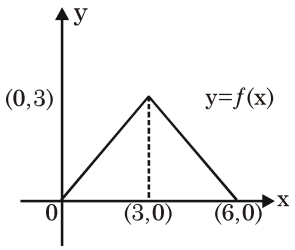
$\therefore$ Apply integrate by part, we get

$= (x^6 + x^2 + 1)\sin(2x) + (3x^5 + x).\cos(2x) + k$

$\therefore \quad f(x) = (x^6 + x^2 + 1)$; $g(x) = (3x^5 + x)$.

11. Ans. (A,C)

$f(x)$ is periodic with period '6'



$$\therefore \int_0^6 f(x) dx = \frac{1}{2} \times 6 \times 3 = 9$$

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12. Ans. (A,C)

$$\int_0^x f(t)\,dt + \int_1^{f(x)} g(t)\,dt = f'(x)$$

Using : $\int_a^b f(x)\,dx + \int_{c=f(a)}^{d=f(b)} f^{-1}(x)\,dx = bd - ac$

$\Rightarrow x.f(x) - 0\times 1 = f'(x)$ (think !)

$\Rightarrow \dfrac{f'(x)}{f(x)} = x \qquad (\because f(x) \neq 0)$

$\Rightarrow \ell n f(x) = \dfrac{x^2}{2} + C$

put $x = 0 \Rightarrow C = 0$

$\Rightarrow f(x) = e^{\frac{x^2}{2}}$

$f(\sqrt{2}) = e \Rightarrow g(e) = \sqrt{2}$

$f(1) = \sqrt{e}$

13. Ans. (B)

(P) $\lim\limits_{n\to\infty} \int_0^2 \left(1+\dfrac{t}{n+1}\right)^n dt$

$$\frac{\left[\left(1+\frac{t}{n+1}\right)^{n+1}\right]_0^2}{(n+1)\left(\frac{1}{n+1}\right)}$$

$\lim\limits_{n\to\infty}\left(1+\dfrac{2}{n+1}\right)^{n+1} - 1 = \boxed{e^2-1}$ $\boxed{P\to 3}$

(Q) $f'(x) = f(x)$, $f(0) = 1$

$f(x) + g(x) = x^2$

$f(x) = e^x$

$g(x) = x^2 - e^x$

$\int_0^1 e^x(x^2 - e^x)\,dx$

$\int_0^1 (x^2e^x - e^{2x})\,dx$

$\left[x^2e^x - 2\int xe^x - \dfrac{e^{2x}}{2}\right]_0^1$

$\left[x^2e^x - 2\left(xe^4 - e^x\right) - \dfrac{e^{2x}}{2}\right]_0^1$

$\left(e - 2(e-e) - \dfrac{e^2}{2}\right) - \left(2 - \dfrac{1}{2}\right)$

$e - \dfrac{e^2}{2} - \dfrac{3}{2}$ $\boxed{Q\to 1}$

(R) $\int_0^1 e^{e^x}(1+xe^x)\,dx$

$e^x = t \Rightarrow e^x dx = dt \Rightarrow dx = \dfrac{dt}{t}$

$\int_1^e e^t(1+\ell nt.t)\dfrac{dt}{t}$

$\int_1^e e^t\left(\ell nt + \dfrac{1}{t}\right)dt$

$\left[e^t\ \ell nt\right]_1^e \Rightarrow \boxed{e^e - 0}$ $\boxed{R\to 4}$

(S) $\lim\limits_{K\to 0}\dfrac{1}{K}\int_0^K (1+\sin 2x)^{\frac{1}{x}}\,dx$

$\lim\limits_{K\to 0}\left((1+\sin 2K)^{\frac{1}{K}}\right)$

$e^{\lim\limits_{K\to 0}\left(\frac{\sin 2K}{2K}\right)^2} = e^2$ $\boxed{S\to 2}$

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SECTION–III

1. Ans. 2

Sol. $\alpha = \int_{2017}^{2019} (x-2017)^2 f(x-2017)\,dx$

$x - 2017 = t$

$\alpha = \int_0^2 t^2 f(t)\,dt$

$\beta = \int_{-2019}^{-2015} (x+2017)^2 f(x+2017)\,dx$

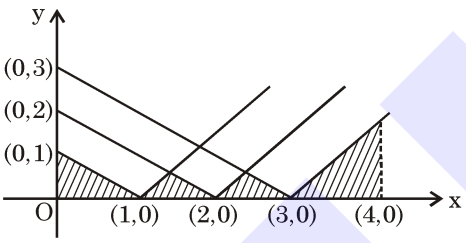
$x + 2017 = z$

$\beta = \int_{-2}^{2} z^2 f(z)\,dz = 2\alpha$

$\frac{\beta}{\alpha} = 2$

2. Ans. 5

Sol.



Required value

$= \frac{1}{2}(1)(1) + 2\left(\frac{1}{2}\times 1\times\frac{1}{2}\right) + \frac{1}{2}\times 1\times 1 = \frac{3}{2}$

3. Ans. 9

4. Ans. 9

Sol. $f(x) = -\int_0^1 f^2(t)dt - 2x\int_0^1 f(t)dt$

$\therefore \quad f'(x) = -2\int_0^1 f(t)dt \qquad .....(i)$

$=$ constant

$\Rightarrow$ f must be linear.

So, let $f(x) = (ax + b)$

$\therefore \quad a = -2\int_0^1 (at+b)dt$

& $\quad b = -\int_0^1 (at+b)^2 dt$

$\Rightarrow \quad a = 3, b = -3$

$\Rightarrow \quad f(x) = 3(x-1)$

So, $f(4) = 3(4-1) = 9$

5. Ans. 2

Sol. $\frac{1}{n\pi}.2n\int_0^1 \frac{dx}{\sqrt{\{x\}\{-x\}}} = \frac{2}{\pi}\int_0^1 \frac{dx}{\sqrt{x(1-x)}}$

$= \frac{4}{\pi}\left(\sin^{-1}\sqrt{x}\right)_0^1 = 2$

6. Ans. 7

Sol. $I_7 = \int_0^1 \underset{I}{\left(1-x^4\right)}.\underset{II}{1}\,dx$

$I_7 = \left[x\left(1-x^4\right)^7\right]_0^1 - \int_0^1 7\left(1-x^4\right)^6\left(-4x^3\right).x\,dx$

$I_7 = 0 + 28\int_0^1 \left(1-x^4\right)^6 x^4 dx$

$I_7 = 28\int_0^1 \left(1-x^4\right)^6\left[1-\left(1-x^4\right)\right]dx$

$I_7 = 28I_6 - 28I_7 \Rightarrow 29I_7 = 28I_6$

$\Rightarrow \frac{29I_7}{4I_6} = 7$

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